

MID-SEMESTER EXAMINATION, November-2023

COMPUTER NETWORKING (CSE 3034)

Programme: B.Tech(CSE, CSIT)
Full Marks: 30

Semester: 5th
Time: 2 Hours

Subject/Course Learning Outcome	*Taxonomy Level	Ques. Nos.	Marks
Able to understand the architectural principles of computer networking and compare different approaches to organizing networks.	L3	1	6
Able to analyze the physical characteristics and functions of the physical devices and interfaces so that transmission can occur in different transmission medium.	L4	2	6
Able to examine various Data Link layer design issues and Data Link protocols.	L5	3,4,5	18

*Bloom's taxonomy levels: Remembering (L1), Understanding (L2), Application (L3), Analysis (L4), Evaluation (L5), Creation (L6)

Answer all questions. Each question carries equal mark.

1. (a) Outline the principal difference between connectionless communication and connection-oriented communication? Give one example of protocol that uses (i) connectionless communication (ii) connection-oriented communication 2
- (b) Explain briefly the essential difference between circuit switching and packet switching? 2
- (c) An image is 1024 x 1024 pixels with 4 bytes/pixel. Assume the image is uncompressed. Apply the basic concept of transmission theory to find the time it takes to transmit it over a 56-kbps modem channel? Over a 10-Mbps Ethernet? 2
2. (a) What are the key components of an optical transmission system? State the advantages of optical fiber over coaxial cable. 2

(b) Explain, why the refractive index of cladding is less than that of core in an optical fiber cable used for transmission of signal. Also state the difference between multimode and single mode fiber. 2

(c) Compare the maximum data rate of a noiseless 4-kHz channel using 2

(a) Analog encoding (using QPSK) with 2 bits per sample.

(b) The T1 PCM system.

3. (a) Explain the different services that the Data Link Layer provides to the Network Layer. 2

(b) The following character encoding is used in a data link protocol: 2

A: 01000111 B: 11100011 FLAG: 01111110 ESC: 11100000

Determine the bit sequence transmitted (in binary) for the four-character frame A ESC B FLAG, when each of the following framing methods is used: (a) Byte count. (b) Starting and ending flag bytes with bit stuffing.

(c) A bit string, 1101011111010111110101111110 needs to be transmitted at the data link layer. Determine the string actually transmitted after bit stuffing. 2

4. (a) Identify and correct the error bit in the received code (110001101), assuming it was originally transmitted using even parity Hamming code (111001101)? 2

(b) 16 bits messages are transmitted using a Hamming code. How many check bits are needed to ensure that the receiver can detect and correct single bit errors? Estimate the bit pattern transmitted for the message 1101001100110101. Assume that even parity is used in the Hamming code. 2

(c) A data bit stream 10101010 is needed to be transmitted using the standard CRC method. The generator polynomial is x^3+x^2+1 . Estimate the actual encoded bit string transmitted. 2

5. (a) Applying the AMI encoding scheme sketch the encoded output for the bit stream: 1110110010110100. 2
- (b) A group of N stations shares a 56-kbps pure ALOHA channel. Each station outputs a 1000-bit frame on an average of once every 200 sec, even if the previous one has not yet been sent (e.g., the stations can buffer outgoing frames). Solve the problem and find the maximum value of N . 2
- (c) Determine the Maximum Throughput that occurs in case of pure ALOHA. 2

End of Questions